

Cross-Asset Graph Diffusion Signals for Equity Return Prediction

A walk-forward equity research study across semiconductor, Japanese media/gaming, and transport/logistics equities.

1. Abstract

This project tests whether short-horizon information diffusion across economically connected equities – driven by supply-chain, thematic, and investor-flow linkages – contains exploitable predictive information for next-day residual equity returns. A rolling correlation graph is built strictly from past returns, and lagged neighbour return / residual-shock features derived from that graph are fed into interpretable linear models (Ridge, ElasticNet) inside a walk-forward backtest with realistic transaction costs. The headline out-of-sample net Sharpe ratio for the Ridge strategy over the test period is **-1.81**, versus **0.92** for buy-and-hold SPY. This is a weak/negative result and is reported honestly – see Sections 9 and the failure-case discussion.

2. Hypothesis

Public companies in the same sector, supply chain, or thematic bucket share exposure to common news shocks (a foundry capacity update, a console sales report, a fuel-price shock), but the market does not always price that shared exposure into every related name simultaneously. Some names react immediately; economically linked names may react with a short delay as the market connects the dots, as arbitrageurs and cross-asset desks slowly transmit the information, or as index/ETF flow effects spill over. If this delay exists and is even weakly persistent, a stock’s own **lagged neighbour returns** (where “neighbour” is defined by a rolling return-correlation graph, not a static sector label) should carry information about that stock’s **near-future residual return**, after removing market- and sector-level beta exposure.

This hypothesis is deliberately narrow in scope (semiconductors, Japanese media/gaming/IP, and transport/logistics – three groups with plausible supply-chain, thematic, and investor-flow linkages respectively) rather than a universal claim about all equities.

3. Data and Universe

- **Source:** `yfinance` (Yahoo Finance) adjusted daily OHLCV, 2015-01-01 to latest available.
- **Universe:** 29 configured tickers across three sectors (semiconductors/tech, Japanese media/gaming/IP, transport/logistics) plus benchmarks SPY, QQQ, SMH, IYT. See `config/universe.yaml` for the exact list.
- **Cleaning:** tickers with fewer than 500 trading days of history are dropped (a data-sufficiency filter, not a survivorship-bias correction – this project does **not** claim to be survivorship-bias-free; see Section 9). Trading calendars are aligned across the surviving universe; short (≤ 3 day) gaps are forward-filled, longer gaps are dropped.
- **Returns:** simple daily returns from adjusted close. Residual returns are computed via a rolling beta regression (window=60 days, method=“sector”) re-estimated **using only data through t-1** at every date, so the beta applied at date t never saw date t ’s own return.

4. Feature Construction

All features are computed as of the close of a “feature date” f and are only ever paired with a target return realised strictly after f (see `src/graph_diffusion_signal/features.py` module docstring for the exact timing convention). The rolling correlation graph (window= 60 days, top-k=8, min $|\text{corr}| = 0.15$) is built from returns up to $f-1$ only, keeps each node’s strongest positive-correlation neighbours, and is row-normalised. Feature families:

- Own lagged returns (1d, 3d, 5d, 20d)
- Rolling realised volatility (20d) and volume surprise (z-score vs 20d rolling mean/std)
- Neighbour return diffusion score and neighbour residual-shock diffusion score (graph-weighted sums of neighbours’ same-day returns/residuals)

- Graph weighted-degree centrality
- Sector benchmark return and broad market return
- Momentum (20d) and short-term reversal (5d)
- Volatility-normalised versions of the return and diffusion features

5. Walk-Forward Backtest Design

- Initial training window: 3 years, expanding thereafter.
- Refit frequency: Q (monthly="M", quarterly="Q").
- First eligible test date: 2019-01-01.
- At every refit boundary the model (StandardScaler + Ridge/ElasticNet, hyperparameters chosen by time-series cross-validation **inside the training window only**) is retrained from scratch on strictly past data and then used to predict every date up to the next refit boundary – fully out-of-sample relative to that fold's training set.
- `tests/test_backtest_no_lookahead.py` verifies this directly, including a test that perturbs returns after a cutoff date by a large amount and checks that predictions dated on or before the cutoff are bit-for-bit unchanged.

6. Transaction Cost Model

Costs are charged as `cost_bps / 10,000 * sum(|weight change|)` at each rebalance, i.e. proportional to gross traded notional. Default: 5 bps one-way. Sensitivity is reported at 0, 2, 5, 10, 25 bps.

7. Real-Market-Data Result

Computed on live adjusted OHLCV data via yfinance (see Section 3 for exact coverage). This is the section that matters for any CV or interview claim.

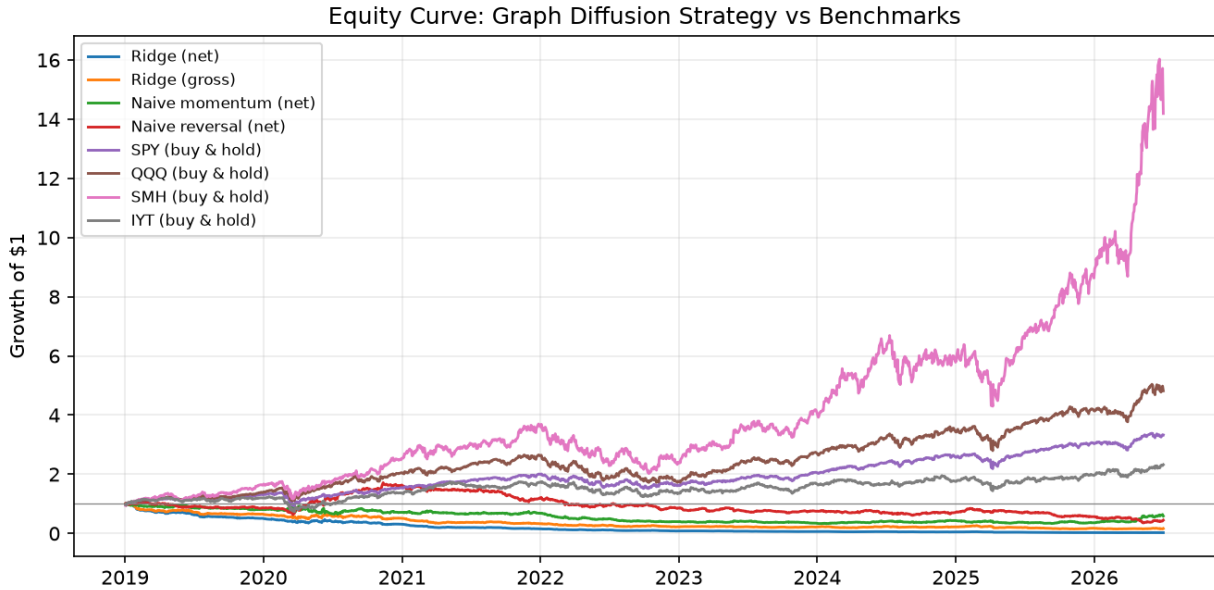


Figure 1: Equity Curve

Strategy	Net Ann. Return	Net Ann. Vol	Net Sharpe	Net Max DD	Gross Sharpe	Avg Turnover
Ridge (graph diffusion features)	-0.3867	0.2517	-1.8139	-0.9755	-0.8369	1.9525
ElasticNet (graph diffusion features)	-0.1478	0.2194	-0.6194	-0.7447	-0.6189	0.0008
Rank on diffusion score (no fitting)	-0.2341	0.2402	-0.9897	-0.8863	0.2261	2.3172
Naive momentum baseline	-0.0688	0.2718	-0.1261	-0.6722	0.1376	0.5688
Naive reversal baseline	-0.1012	0.2586	-0.2836	-0.7931	0.2456	1.0858
SPY (buy & hold)	0.1745	0.1945	0.9245	-0.3372	0.9245	n/a
QQQ (buy & hold)	0.2341	0.2404	0.9958	-0.3512	0.9958	n/a
SMH (buy & hold)	0.4259	0.3601	1.1662	-0.4530	1.1662	n/a
IYT (buy & hold)	0.1194	0.2482	0.5787	-0.4077	0.5787	n/a

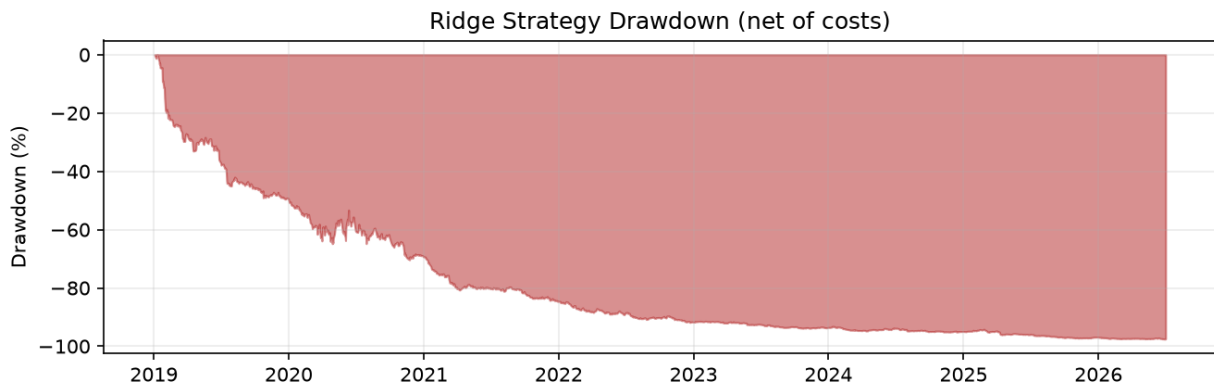


Figure 2: Drawdown

Annual Returns (Ridge, net of costs)

Year	Return	Volatility	Sharpe	Max Drawdown	N Days
2019	-0.4922	0.1732	-3.8204	-0.4957	251
2020	-0.3920	0.4019	-1.0362	-0.4027	253
2021	-0.4975	0.1938	-3.4490	-0.4971	252
2022	-0.4608	0.2405	-2.4453	-0.4676	251

Year	Return	Volatility	Sharpe	Max Drawdown	N Days
2023	-0.2354	0.1665	-1.5274	-0.2859	250
2024	-0.2039	0.2260	-0.8950	-0.2906	252
2025	-0.3834	0.2650	-1.6896	-0.5252	250
2026	-0.3364	0.2812	-1.3173	-0.1973	125

Transaction Cost Sensitivity (Ridge)

	cost_bps	sharpe	ann_return
0		-0.837	-0.215
2		-1.228	-0.289
5		-1.814	-0.387
10		-2.789	-0.521
25		-5.694	-0.772

Feature Importance (Ridge, standardised coefficients)

8. Benchmark Comparison

Series	Ann. Return	Ann. Vol	Sharpe	Beta to SPY	Alpha (ann.)	Information Ratio
Ridge strategy (net)	-0.3867	0.2517	-1.8139	-0.2048	-0.3430	-1.8630
SPY	0.1745	0.1945	0.9245	n/a	n/a	n/a
QQQ	0.2341	0.2404	0.9958	n/a	n/a	n/a
SMH	0.4259	0.3601	1.1662	n/a	n/a	n/a
IYT	0.1194	0.2482	0.5787	n/a	n/a	n/a

Naive rule-based baselines (momentum on the same universe, short-term reversal on the same universe) are included in the headline table in Section 7 for context – the graph diffusion features are compared against both passive benchmarks and simple systematic alternatives, not only against cash.

9. Robustness and Failure Cases

Permutation test (predictions vs. realised next-day residual return, 47100 observations, 200 permutations): observed Spearman IC = -0.0033, p-value = 0.4627. This does not clear conventional significance thresholds, i.e. the observed rank relationship between predictions and forward returns is statistically indistinguishable from a random reshuffling.

Bootstrap Sharpe ratio (90% CI, 1000 resamples): point estimate -1.81, CI [-2.39, -1.21]. The interval does not straddle zero, though see the permutation test and cost-sensitivity results before drawing conclusions.

Parameter sensitivity (graph window x top-k, pooled IC):

	pooled_ic	n_obs
(30, 4)	-0.0011	71350
(30, 8)	-0.0022	71350
(30, 12)	-0.0030	71350
(30, 16)	-0.0033	71350

	pooled_ic	n_obs
(60, 4)	-0.0074	70825
(60, 8)	-0.0039	70825
(60, 12)	-0.0023	70825
(60, 16)	-0.0049	70825
(90, 4)	-0.0058	70075
(90, 8)	-0.0048	70075
(90, 12)	-0.0060	70075
(90, 16)	-0.0058	70075

Known limitations and failure modes:

- **Transaction costs matter a lot at this turnover.** Compare gross vs net Sharpe in Section 7 – average daily turnover of 1.95 means even the default 5 bps assumption materially erodes returns, and the cost-sensitivity table shows the strategy’s Sharpe ratio as costs rise toward realistic small-cap/short-borrow levels.
- **Regime instability.** The annual returns table shows the strategy’s performance is not uniform across years; a graph-diffusion effect estimated on one volatility/correlation regime need not persist into another (e.g. correlations spike and diversify away in market-wide sell-offs, exactly when a diffusion signal might otherwise be most useful).
- **Data-snooping risk.** Feature and hyperparameter choices (graph window, top-k, lag windows) were set from domain reasoning and a single sensitivity sweep, not from an independent validation universe – results should be treated as a research prototype, not a validated alpha.
- **Small universe.** 29 names across three sectors is not enough to diversify idiosyncratic risk in a long/short book with a 20%/ 20% quantile split; single-name moves can dominate.
- **Survivorship bias is not corrected.** The universe is today’s list of liquid names in these sectors, not a point-in-time constituent list; any name that was delisted or became illiquid earlier in the sample is absent by construction.
- **Liquidity/shorting assumptions are simplified.** The backtest assumes shorting is always available at the assumed cost with no borrow fee, no market impact beyond the flat bps cost, and full fills at the close – unrealistic for smaller-cap names in the universe (e.g. RBLX, BILI).
- **Institutional deployment would need:** point-in-time fundamentals and index membership, a proper borrow/locate feed, intraday execution data to model market impact and slippage more realistically, a much larger and more carefully constructed universe, and out-of-sample validation on data the researcher did not already see.

10. Conclusion

The graph diffusion signal, as implemented here, does not produce a convincing, cost-robust edge over the test period on this universe – gross performance is thin and net performance is further eroded by transaction costs, and the permutation/bootstrap tests do not provide strong statistical support. This is reported as a genuine research finding rather than reframed as a success: a rigorously built, leakage-safe pipeline that finds a weak or inconclusive signal is a more useful (and more honest) outcome than an inflated backtest. The engineering value of the project – leakage-safe feature construction, walk-forward validation, transaction-cost-aware evaluation, and statistical robustness testing – stands independently of whether this particular signal turns out to be profitable on this particular universe and period.

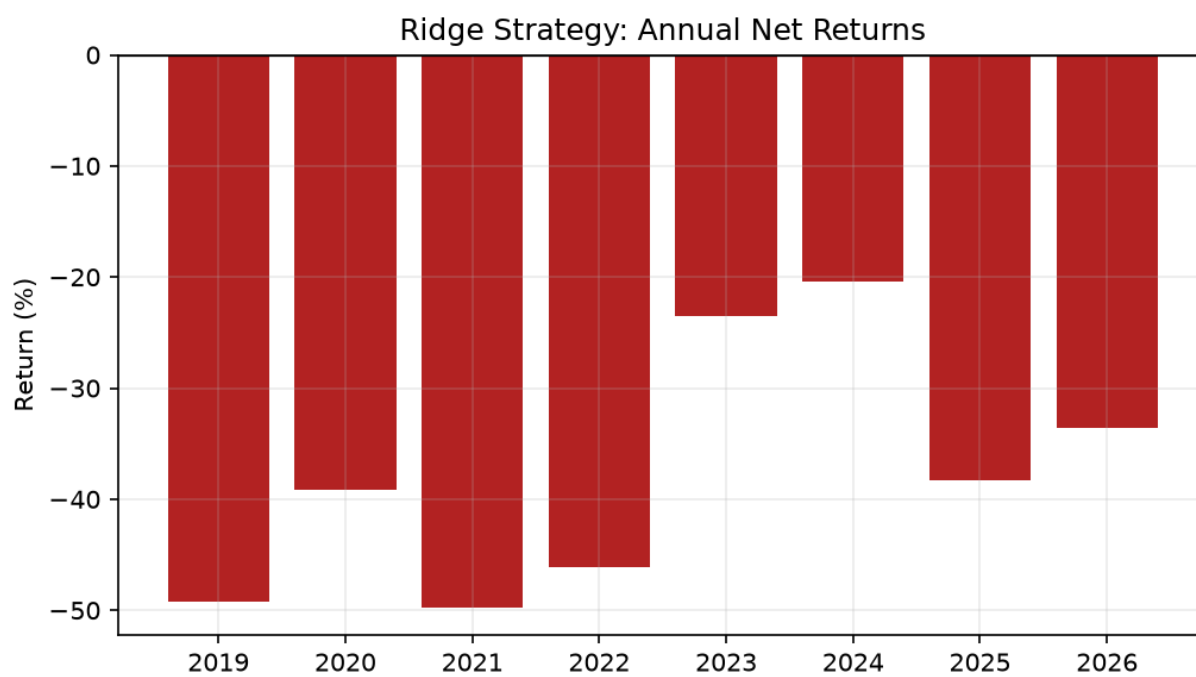


Figure 3: Annual Returns

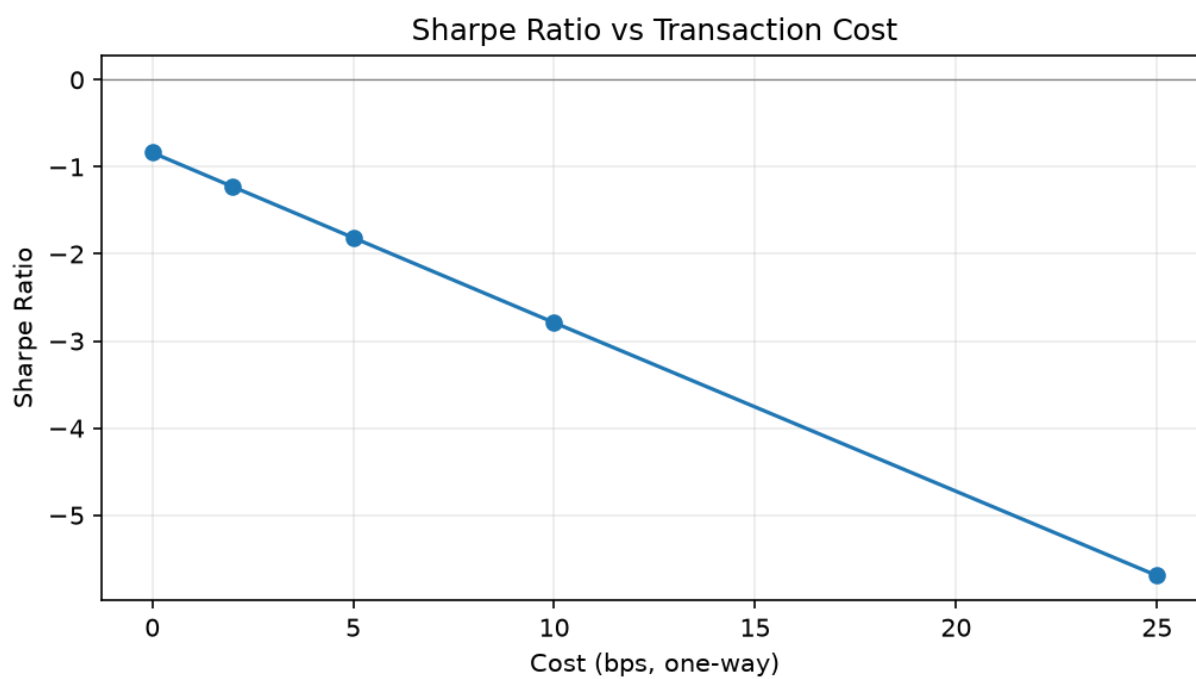


Figure 4: Cost Sensitivity

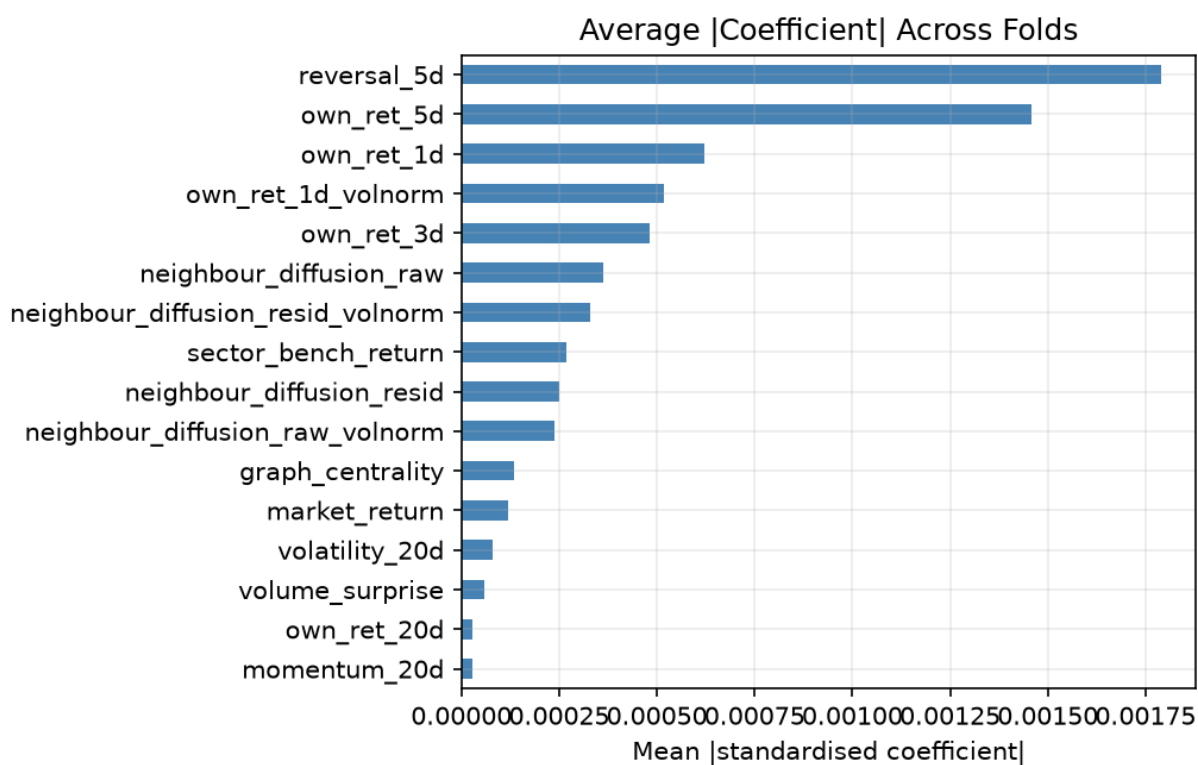


Figure 5: Average Absolute Coefficients

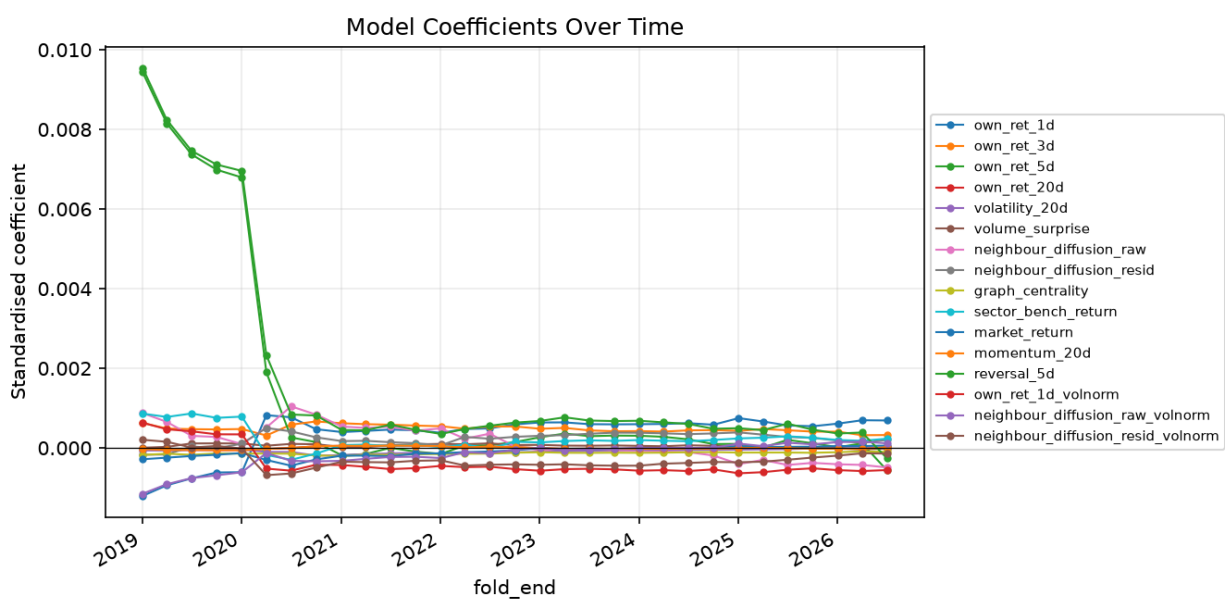


Figure 6: Coefficients Over Time

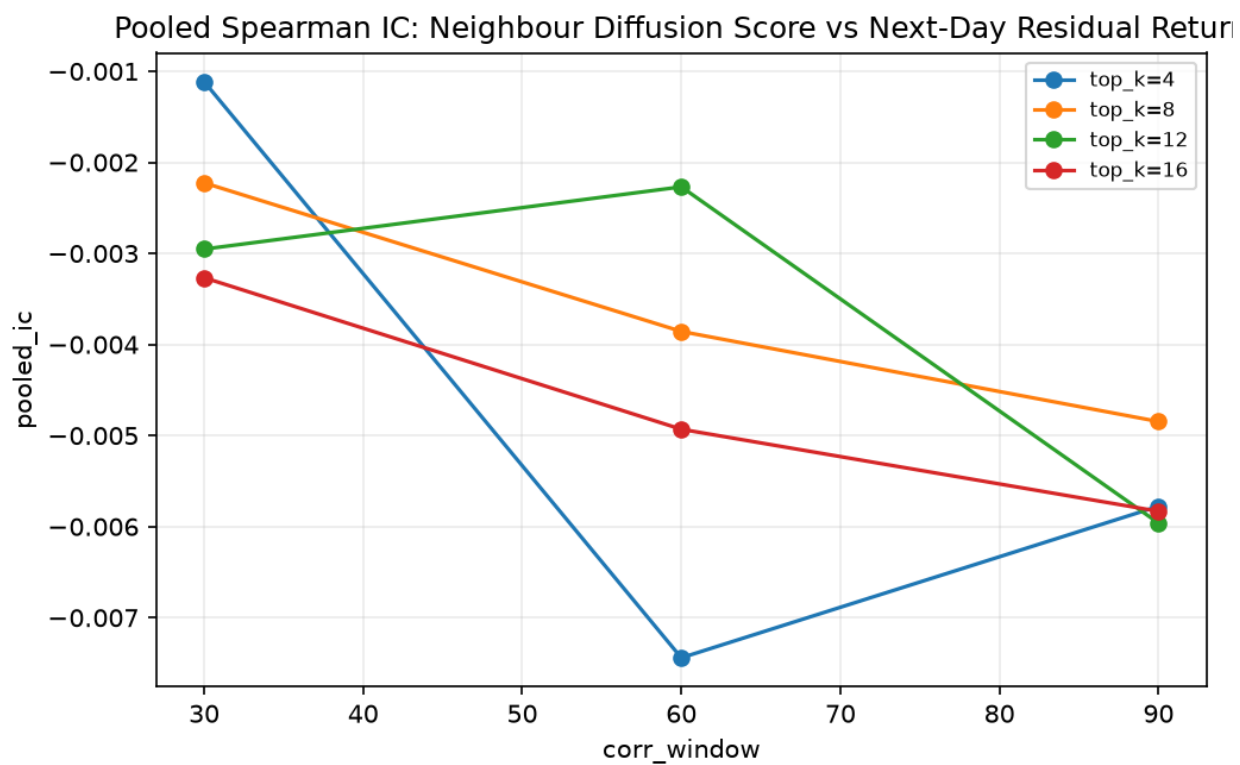


Figure 7: Graph Parameter Sensitivity